

Experiment 5 : Job Pipeline - Jenkins

Aim :

To Build the pipeline of jobs using Maven /Gradle /Ant in Jenkins,create a pipeline script to Test and deploy an application over the tomcat server.

Theory :

Jenkins is an open-source automation server used to implement **Continuous Integration (CI)** and **Continuous Deployment (CD)** in software development. It helps automate tasks like building, testing, and deploying code, improving efficiency and reducing human errors.

Key Concepts:

1. **Automated Builds:** Jenkins automatically compiles code whenever changes are made, ensuring that the software is always in a buildable state.
2. **Continuous Integration:** Developers frequently commit code to a shared repository. Jenkins automatically triggers builds and runs tests to detect issues early.
3. **Pipelines:** Jenkins workflows are defined using **Pipelines** in a `Jenkinsfile`, which can be scripted or declarative.
4. **Plugins:** Jenkins supports plugins for version control (Git), build tools (Maven, Gradle), deployment (Docker, Kubernetes), and more.
5. **Freestyle Projects vs. Pipelines:**
 - **Freestyle Projects:** Simple, GUI-based jobs for basic automation.
 - **Pipelines:** Code-defined workflows for complex CI/CD tasks.

Example of a Simple Jenkins Pipeline (Declarative):

```
pipeline {
    agent any
    stages {
        stage('Build') {
            steps {
                sh 'echo "Building the project"'
            }
        }
        stage('Test') {
            steps {
                sh 'echo "Running tests"'
            }
        }
        stage('Deploy') {
            steps {
                sh 'echo "Deploying the application"'
            }
        }
    }
}
```

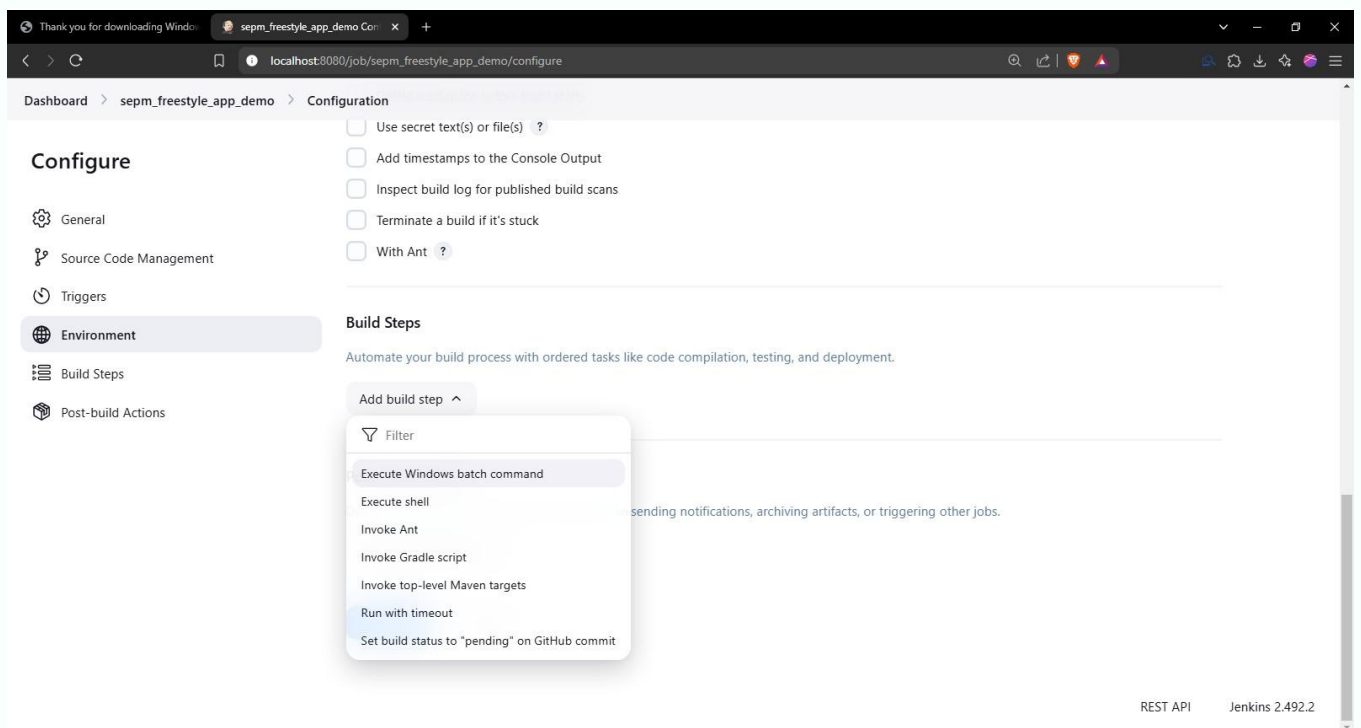
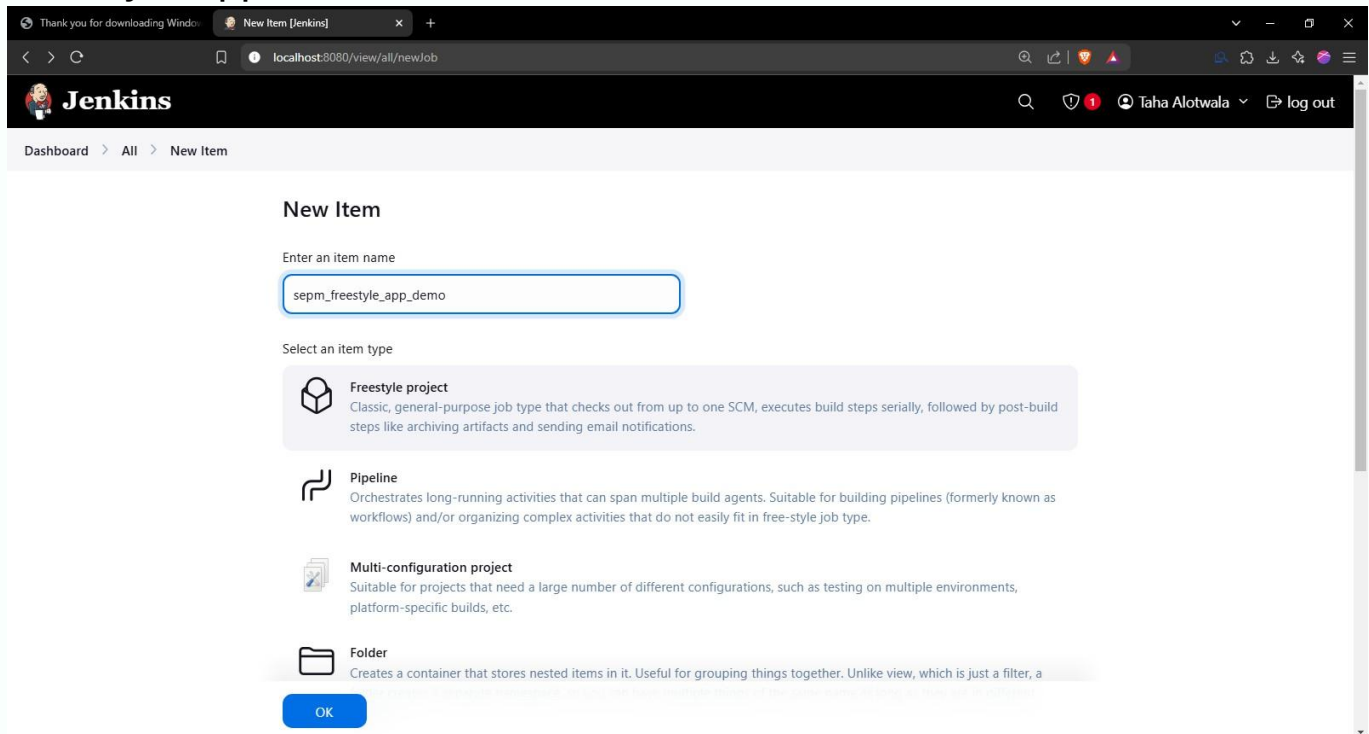
This pipeline defines three stages: **Build**, **Test**, and **Deploy**, each with its own commands.

Benefits of Jenkins in Programming:

- Automates repetitive tasks (builds, tests, deployments).
- Detects bugs early with continuous testing.
- Speeds up the development lifecycle.

Output :

1. Freestyle App



Thank you for downloading Windows
sepm_freestyle_app_demo Con

localhost:8080/job/sepm_freestyle_app_demo/configure

Configure

General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions

☐ Use secret text(s) or file(s) ?
☐ Add timestamps to the Console Output
☐ Inspect build log for published build scans
☐ Terminate a build if it's stuck
☐ With Ant ?

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

[See the list of available environment variables](#)

```
net stats workstation | find "Statistics since"
```

Advanced ▾

Save Apply

Thank you for downloading Windows
sepm_freestyle_app_demo #1

localhost:8080/job/sepm_freestyle_app_demo/1/

Jenkins

Taha Alotwala log out

Dashboard > sepm_freestyle_app_demo > #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

✓ #1 (Apr 2, 2025, 11:43:13 AM)

Started by user [Taha Alotwala](#)

This run spent:

- 12 ms waiting;
- 0.37 sec build duration;
- 0.38 sec total from scheduled to completion.

No changes.

Add description

Keep this build forever

Started 4.9 sec ago
Took [0.37 sec](#)

REST API Jenkins 2.492.2

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sepm_freestyle_app_demo #1

localhost:8080/job/sepm_freestyle_app_demo/1/console

Jenkins

Taha Alotwala log out

Dashboard > sepm_freestyle_app_demo > #1 > Console Output

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#1'
- Timings

Console Output

Download Copy View as plain text

```
Started by user Taha Alotwala
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\sepm_freestyle_app_demo
[sepm_freestyle_app_demo] $ cmd /c call C:\Windows\TEMP\jenkins10778809563619165675.bat

C:\ProgramData\Jenkins\.jenkins\workspace\sepm_freestyle_app_demo>net stats workstation | find "Statistics since"
Statistics since 13-03-2025 04:33:05

C:\ProgramData\Jenkins\.jenkins\workspace\sepm_freestyle_app_demo>exit 0
Finished: SUCCESS
```

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2. Taking Parameters

script :

```
MSYS:/d/MiscRepos/sepm_lab
```

```
1 @echo off
2 echo "Hello %1... Your address is %2"
```

```
example1.cmd[+] [unix] (05:29 01/01/1970) 2,38 All
-- INSERT --
```

Executing the script (unsuccessful) :

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

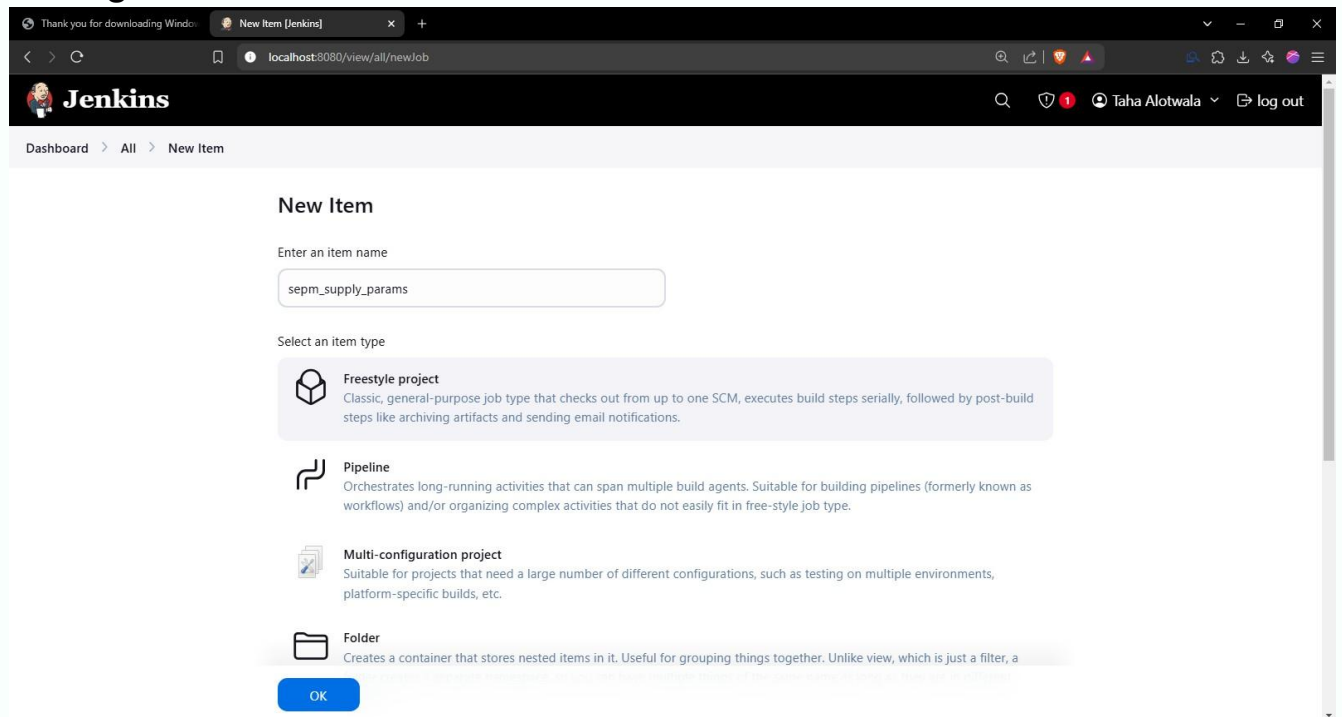
C:\Users\Admin>cd "D:\MiscRepos\sepm_lab\files"

C:\Users\Admin>example1.cmd
'example1.cmd' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Admin>example1.cmd Taha
'example1.cmd' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Admin>
```

Configure Jenkins :



Thank you for downloading Windows
sepm_supply_params Config [x] +

localhost:8080/job/sepm_supply_params/configure

Dashboard > sepm_supply_params > Configuration

☐ With Ant ?

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

≡ **Execute Windows batch command** ?

Command

[See the list of available environment variables](#)

```
D:\MiscRepos\sepm_lab\files\example1.cmd Taha Mazgaon
```

Advanced ▾

Add build step ▾

Post-build Actions

Refer to the help page for a full complete list of available post-build actions and their configuration options.

Save Apply

Thank you for downloading Windows
sepm_supply_params #1 [Jenkins] x +

localhost:8080/job/sepm_supply_params/1/

Jenkins

Dashboard > sepm_supply_params > #1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

✓ #1 (Apr 2, 2025, 12:01:38 PM)

Add description Keep this build forever

Started by user [Taha Alotwala](#)

Started 55 sec ago
Took 0.29 sec

This run spent:

- 6 ms waiting;
- 0.29 sec build duration;
- 0.29 sec total from scheduled to completion.

</> No changes.

REST API Jenkins 2.492.2

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `localhost:8080/job/sepm_supply_params/1/console`. The Jenkins logo and user profile (Taha Alotwala) are visible in the top navigation bar. The breadcrumb trail shows the path: `Dashboard > sepm_supply_params > #1 > Console Output`. On the left sidebar, the 'Console Output' option is selected. The main content area displays the console output with a green checkmark icon and the title 'Console Output'. Above the output text are buttons for 'Download', 'Copy', and 'View as plain text'. The console output text is as follows:

```
Started by user Taha Alotwala
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\sepm_supply_params
[sepm_supply_params] $ cmd /c call C:\Windows\TEMP\jenkins15807160200406300302.bat

C:\ProgramData\Jenkins\.jenkins\workspace\sepm_supply_params>D:\MiscRepos\sepm_lab\files\example1.cmd Taha Mazgaon
"Hello Taha... Your address is Mazgaon"
Finished: SUCCESS
```

At the bottom right of the interface, it says 'REST API' and 'Jenkins 2.492.2'.

3. Running a Java Program

The code :

The screenshot shows a code editor with a file named `JavaInJenkins.java`. The code is as follows:

```
6 class JavaInJenkins
5 {
4     public static void main(String[] args)
3     {
2         System.out.println("Hello, from the java file");
1     }
7 }
```


Running the file through the terminal :

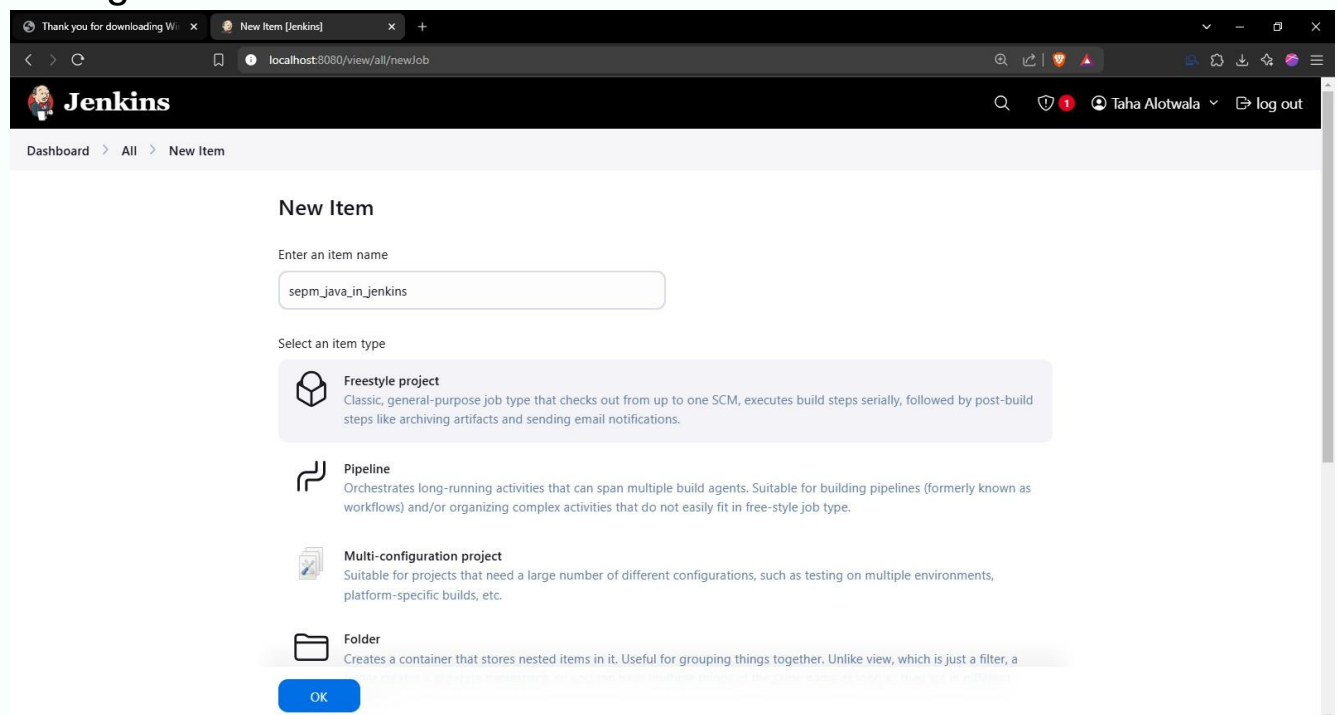
```
/d/MiscRepos/sepm_lab/files git:(master)±5 (5.232s)
vi JavaInJenkins.java

/d/MiscRepos/sepm_lab/files git:(master)±5 (1.298s)
javac JavaInJenkins.java

/d/MiscRepos/sepm_lab/files git:(master)±5 (1.296s)
java JavaInJenkins.java
Hello, from the java file

/d/MiscRepos/sepm_lab/files git:(master)✓ ±5
|
```

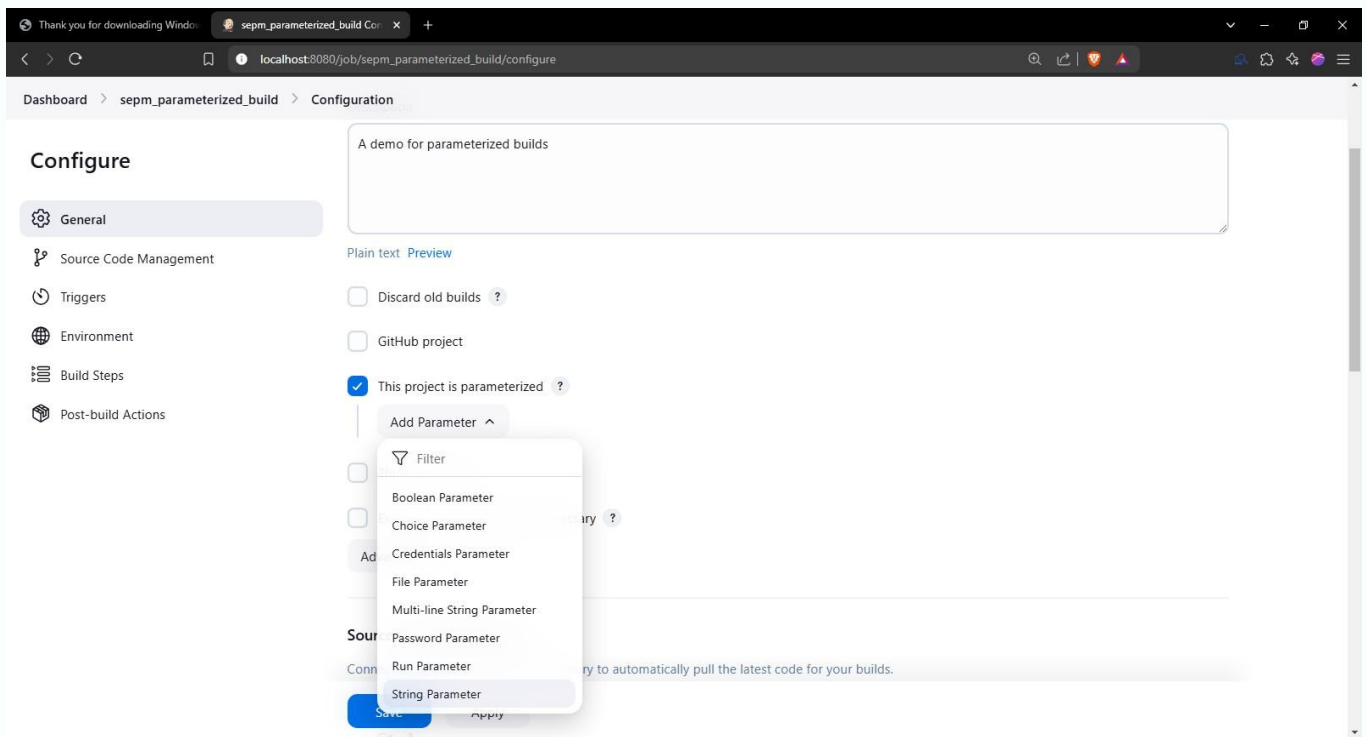
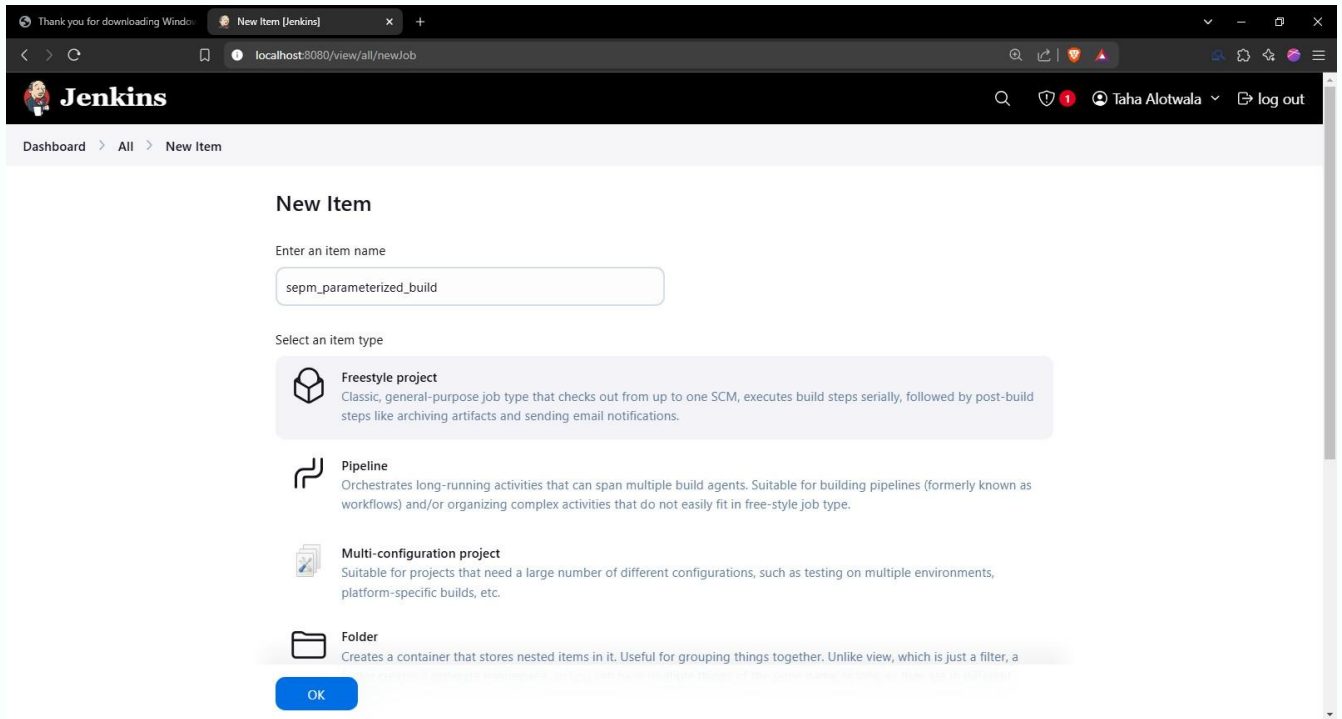
Configure Jenkins :

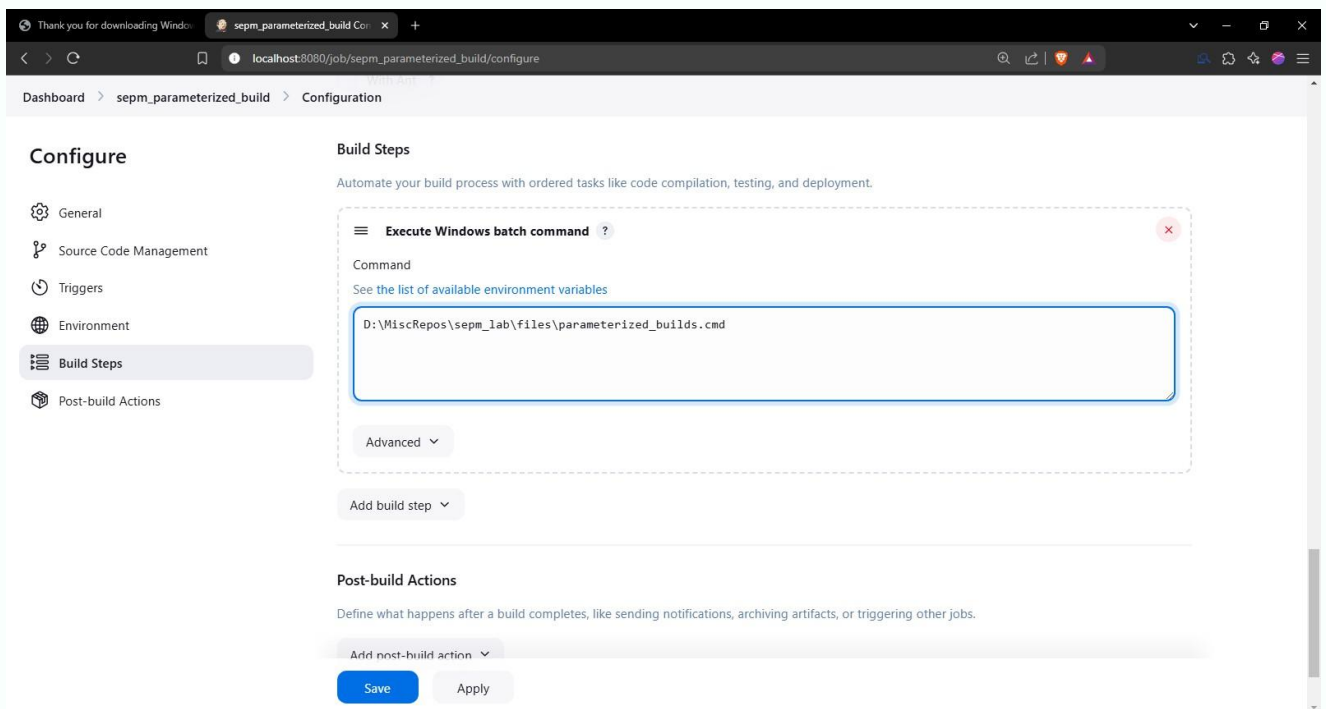
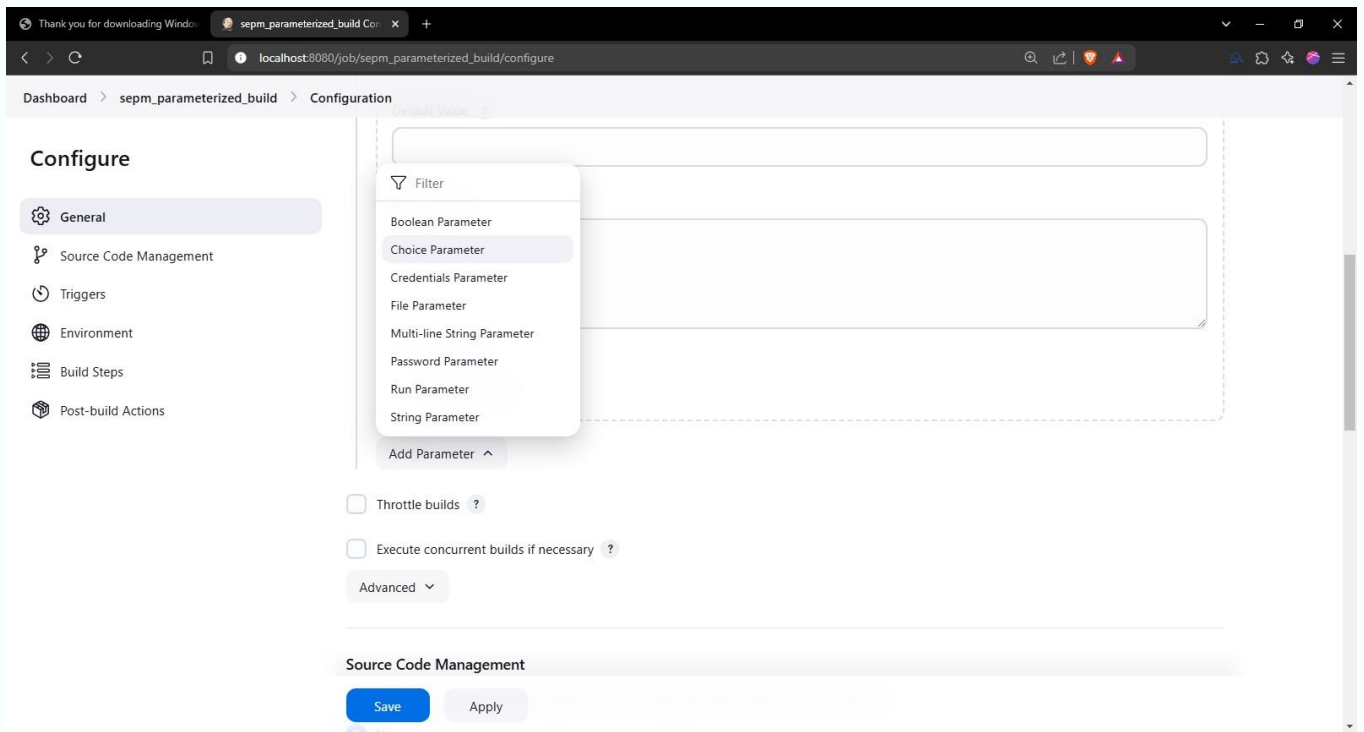


The screenshot shows the Jenkins Configuration page for a job named 'sepm_java_in_jenkins'. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'Build Steps' tab is selected. The main content area is titled 'Build Steps' and includes a description: 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' Below this, there is a section for 'Execute Windows batch command' with a text area containing the following commands: `javac D:\MiscRepos\sepm_lab\files\JavaInJenkins.java` and `java D:\MiscRepos\sepm_lab\files\JavaInJenkins.java`. There is an 'Advanced' dropdown and an 'Add build step' button. Below the build steps section, there is a 'Post-build Actions' section with a description: 'Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.' It includes an 'Add post-build action' dropdown and 'Save' and 'Apply' buttons.

The screenshot shows the Jenkins Console Output page for a job named 'sepm_java_in_jenkins'. The left sidebar contains a 'Console Output' section with a list of tabs: Status, Changes, Console Output, Edit Build Information, Delete build '#1', and Timings. The 'Console Output' tab is selected. The main content area is titled 'Console Output' and includes a green checkmark icon. Below this, there is a text area containing the following output: `Started by user Taha Alotwala`, `Running as SYSTEM`, `Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\sepm_java_in_jenkins`, `[sepm_java_in_jenkins] $ cmd /c call C:\Windows\TEMP\jenkins13007954911449735543.bat`, `C:\ProgramData\Jenkins\jenkins\workspace\sepm_java_in_jenkins>javac D:\MiscRepos\sepm_lab\files\JavaInJenkins.java`, `C:\ProgramData\Jenkins\jenkins\workspace\sepm_java_in_jenkins>java D:\MiscRepos\sepm_lab\files\JavaInJenkins.java`, `Hello, from the java file`, `C:\ProgramData\Jenkins\jenkins\workspace\sepm_java_in_jenkins>exit 0`, and `Finished: SUCCESS`. There are buttons for 'Download', 'Copy', and 'View as plain text'. At the bottom right, there is a 'REST API' link and the version 'Jenkins 2.492.2'.

4. Parameterized Builds





Thank you for downloading Windows Jenkins

localhost:8080/job/sepm_parameterized_build/build?delay=0sec

Jenkins

Taha Alotwala log out

Dashboard > sepm_parameterized_build >

- Status
- Changes
- Workspace
- Build with Parameters
- Configure
- Delete Project
- Rename

Project sepm_parameterized_build

This build requires parameters:

name
Taha

city
Mumbai

Build Cancel

Builds

No builds

Jenkins 2.492.2

```
.. parameterized_builds.cmd
1 @echo off
2 echo "Hello your name is : %name%, and your city is %city%."
```

Thank you for downloading Windows sepm_parameterized_build #2

localhost:8080/job/sepm_parameterized_build/2/console

Jenkins

Taha Alotwala log out

Dashboard > sepm_parameterized_build > #2 > Console Output

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#2'
- Parameters
- Timings
- Previous Build

Console Output

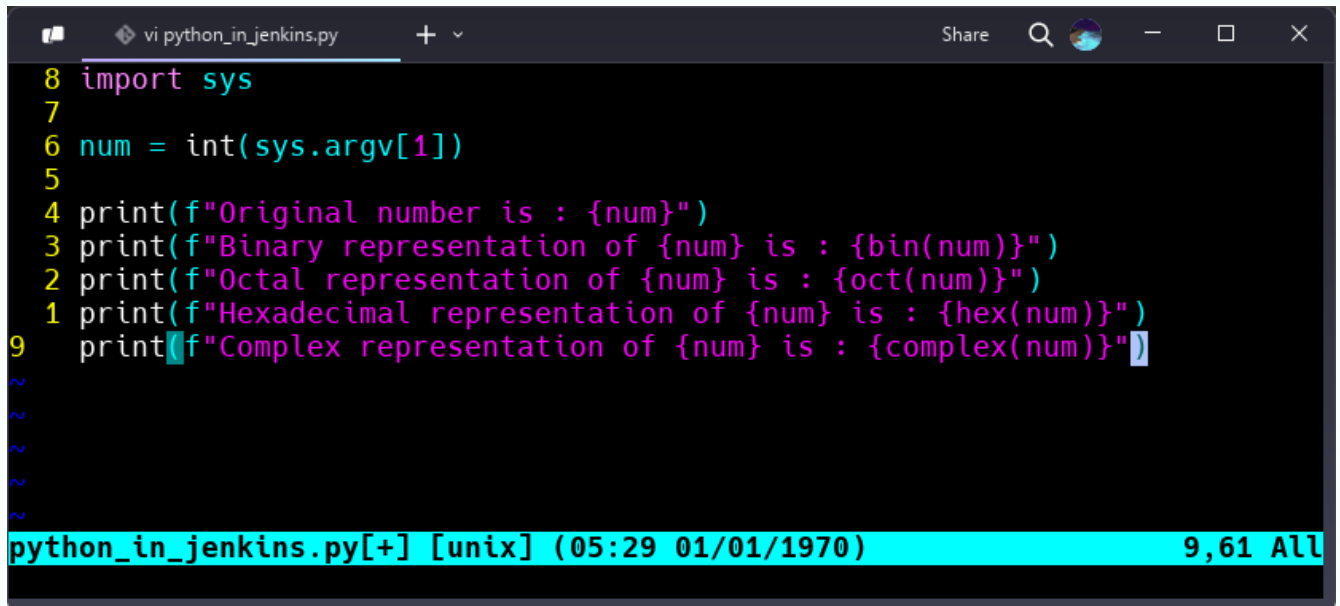
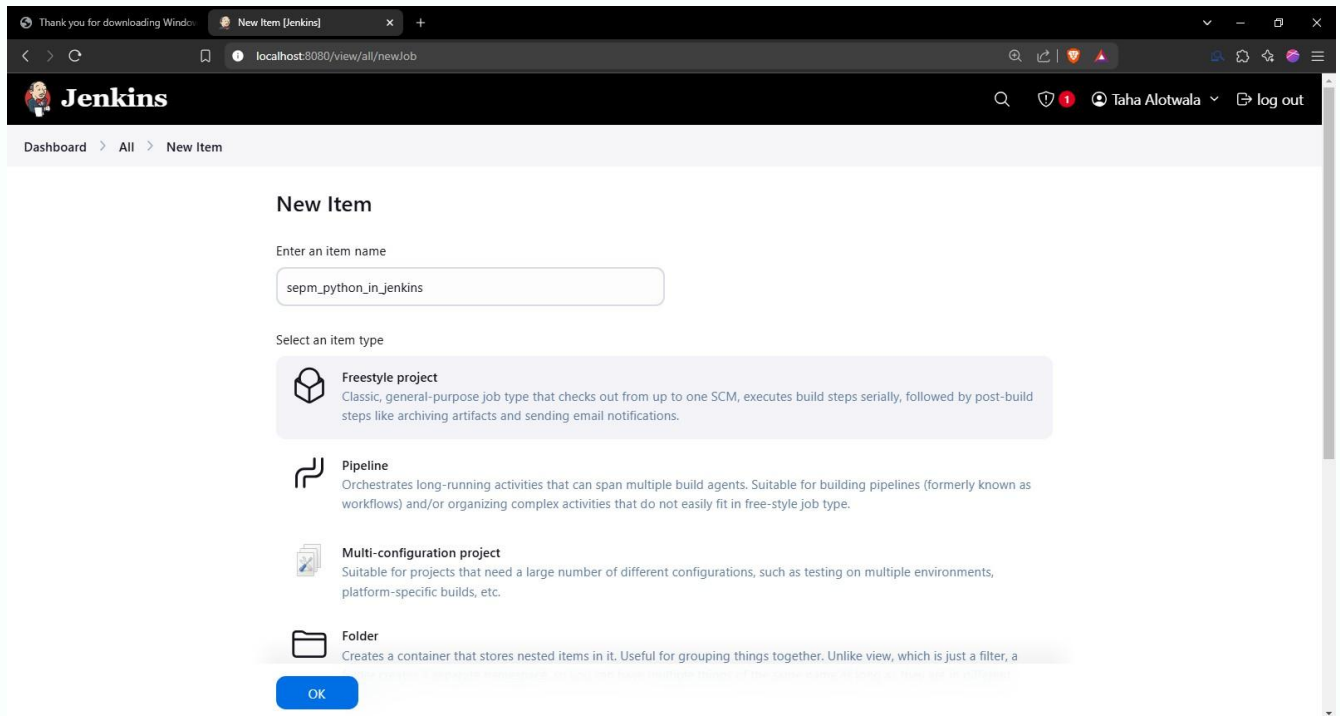
Download Copy View as plain text

Started by user Taha Alotwala
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\sepm_parameterized_build
[sepm_parameterized_build] \$ cmd /c call C:\Windows\TEMP\jenkins2249229840313057984.bat

C:\ProgramData\Jenkins\jenkins\workspace\sepm_parameterized_build>D:\MiscRepos\sepm_lab\files\parameterized_builds.cmd
"Hello your name is : Taha, and your city is Mumbai."
Finished: SUCCESS

REST API Jenkins 2.492.2

5. Running a python script

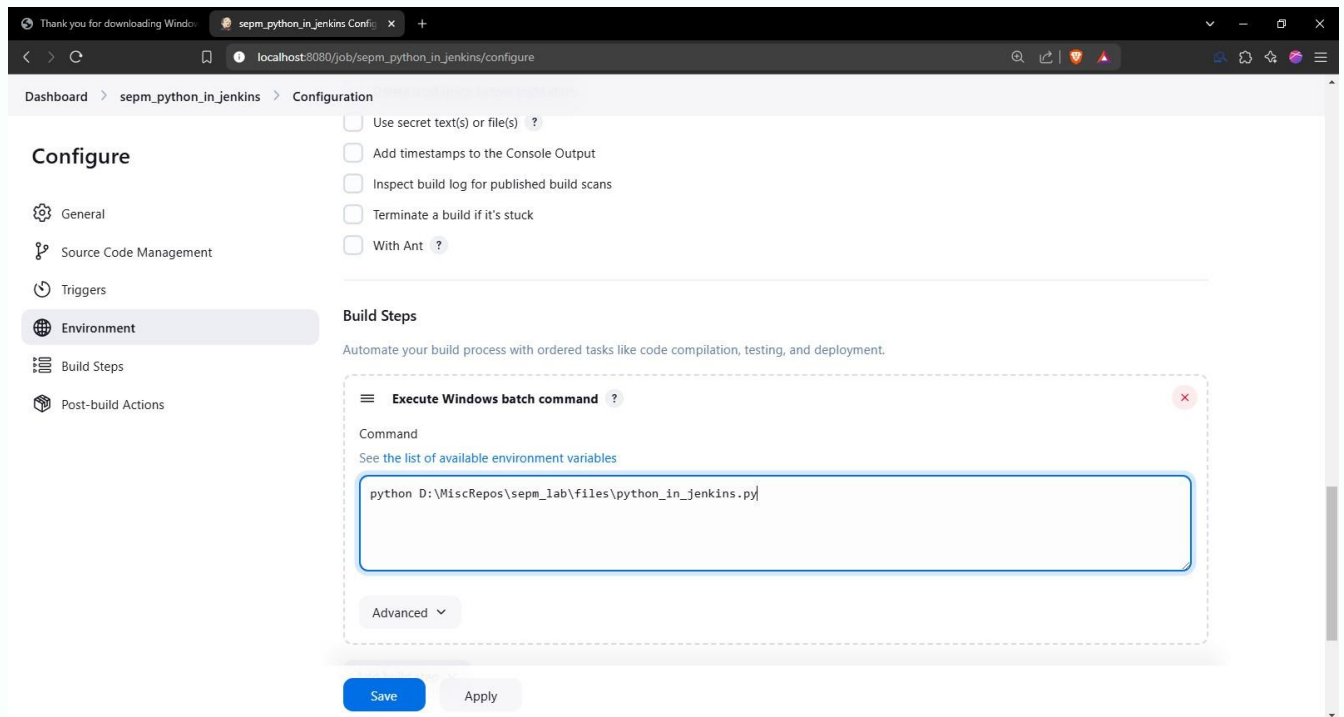


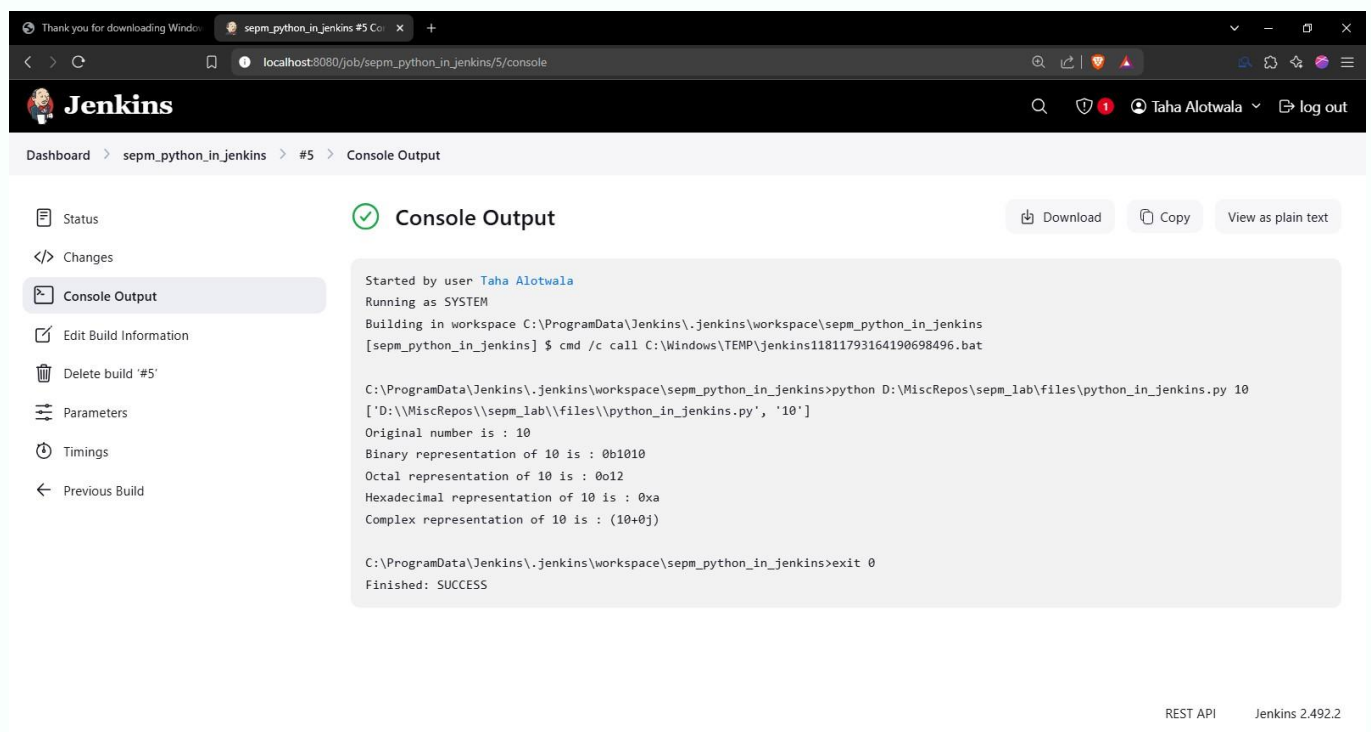
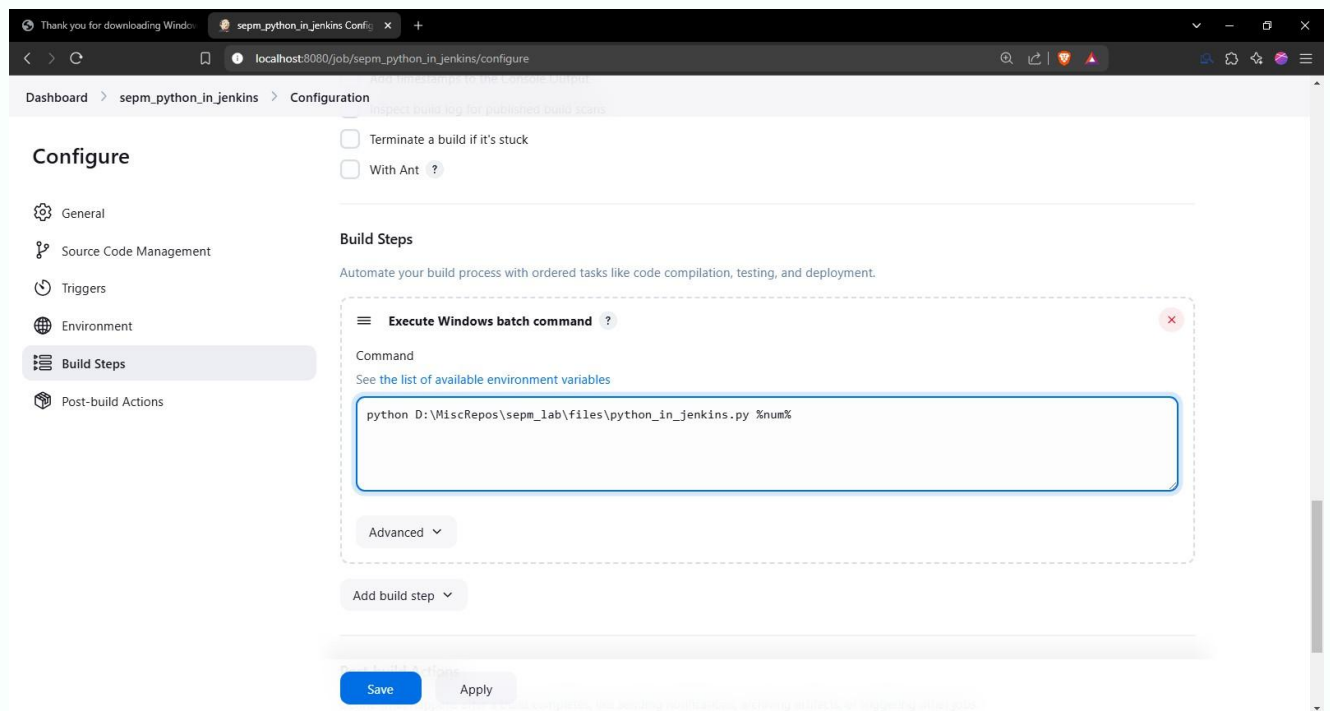
```
../MiscRepos/sepm_lab/files + v
Share 🔍 - □ ×

/d/MiscRepos/sepm_lab/files git:(master)±5 (4m 5.85s)
vi python_in_jenkins.py

/d/MiscRepos/sepm_lab/files git:(master)±5 (0.612s)
python python_in_jenkins.py 10
Original number is : 10
Binary representation of 10 is : 0b1010
Octal representation of 10 is : 0o12
Hexadecimal representation of 10 is : 0xa
Complex representation of 10 is : (10+0j)

/d/MiscRepos/sepm_lab/files git:(master) v ±5
```





Conclusion :

Thus, we have successfully studied Continuous Integration and installed, configured, and understood programming with Jenkins.